Started on	Sunday, 12 February 2023, 8:41 PM
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Time taken	10 secs
Marks	0/41
Grade	**0** out of 100

Question **1**

Not answered

Marked out of 1

A graphical display of a frequency distribution that uses classes and vertical bars of various heights to represent the frequencies is known as

Select one:

- ROC Curve
- Forest plot
- Histogram
- Parabola
- Pie chart

Your answer is incorrect.

A histogram is similar to a bar chart but has no gaps between the bars as the data are continuous (attached image). The width of each bar of the histogram relates to a range of values for the variable.

The correct answer is: Histogram

Question 2

Not answered

Marked out of 1

When straight-line segments are connected through the midpoints at the top of the rectangles of a histogram with the two ends tied down to the horizontal axis, the resulting graph is called

Select one:

- A frequency polygon
- Pie chart
- ROC curve
- Bar cart
- Forest plot

Your answer is incorrect.

A frequency polygon can be constructed by joining the midpoints of the tops of the bars in a histogram.

The correct answer is: A frequency polygon

Question 3

Not answered

Marked out of 1

A box-whisker plot depicts all of the following in graphical form except

Select one:

- Median
- Upper quartile
- Lowest value
- Standard deviation
- Lower quartile

Your answer is incorrect.

See the attached image. Standard deviation is not associated with median values that are usually used in drawing box-whisker plots.

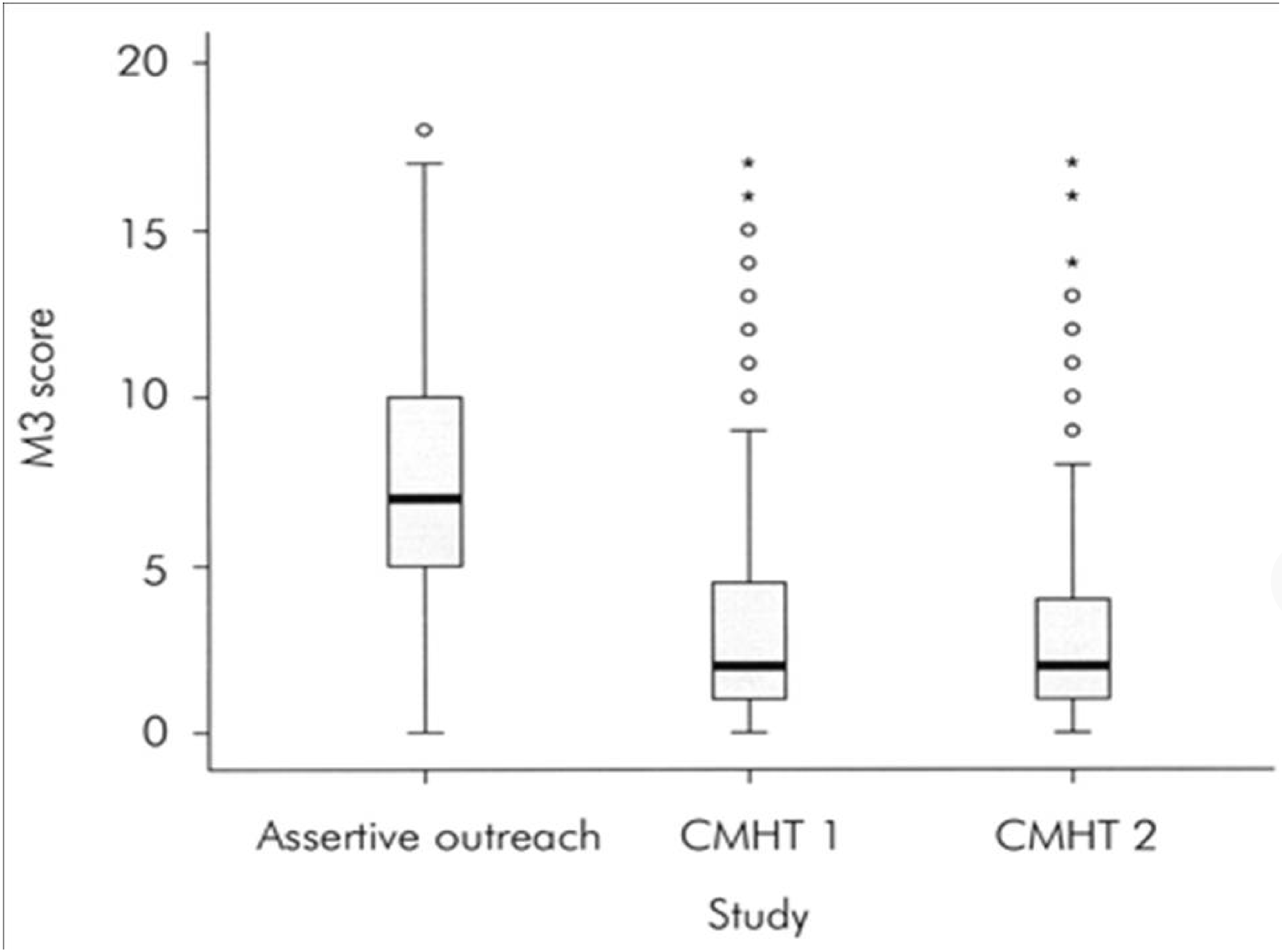
The correct answer is: Standard deviation

Question 4

Not answered

Marked out of 1

The attached image shows a comparison of three teams: an assertive outreach team and two community mental health teams using Severity and risk score of Matching Resources to Care scale (M3). Which of the following inference is correct


20
15
10
5
0
M3 score
Assertive outreach
CMHT 1
CMHT 2
Study

Select one:

- CMHT1 has a symmetrical distribution of scores
- The rectangular boxes contain 25% of observations
- CMHT1 has more patients with high severity and risk than the Assertive outreach team
- The assertive outreach team has no outliers
- Scores that are outlying the range for CMHTs are still within the 95% range for the assertive outreach team

Your answer is incorrect.

If the whiskers are approximately the same length, the distribution of the data values will be approximately symmetrical. This is not true with respect to CMHT. The assertive outreach team has at least one outlier. The rectangular boxes contain observations between 1st and last quartile; hence they have 50% of data. If the median is close to the center of the box, the distribution of the data values will be approximately symmetrical. Schneider, J., Brandon, T., Wooff, D., et al. (2006) Assertive Outreach: policy and reality. Psychiatr Bull, 30, 89-94.

The correct answer is: Scores that are outlying the range for CMHTs are still within the 95% range for the assertive outreach team

Question 5

Not answered

Marked out of 1

All of the following processes could be applied to convert certain types of data in a skewed distribution to a normal distribution except

Select one:

- Reciprocal transformation
- Square root transformation
- Square transformation
- Logistic regression
- Logit transformation

Your answer is incorrect.

As sampled data may not always satisfy Normal distribution - which is a requirement for many robust parametric tests, we may have to convert our raw data into transformed data by taking the same mathematical transformation of each observation. Various methods listed above are used - except logistic regression which is not a transformation method.

The correct answer is: Logistic regression

Question 6

Not answered

Marked out of 1

Scores on a new psychometric instrument that measures working memory capacity are noted to be normally distributed in a large sample of university students. If the mean score is 50 units, with the standard deviation being 10, what percentage of university students will score more than 60?

Select one:

- 0.68
- 0.32
- 0.16
- 0.05
- 0.95

Your answer is incorrect.

The score 60 is one SD higher than the mean score. In any normal distribution curve, 68% will fall between -1 and +1 SD. Of the remaining 32%, 16% will be below -1SD and 16% will be above +1SD, i.e., the value 60.

The correct answer is: 0.16

Question 7

Not answered

Marked out of 1

Which of the following statement is true regarding positively skewed distribution?

Select one:

- Tail of the distribution is to the left of the mean
- Most data values will fall to the left of the mean
- Mean is to the left of the median
- Mode is to the right of the median
- Outliers tend to decrease the value of mean

Your answer is incorrect.

In a positively skewed distribution, most of the data will fall to the left of the mean, while the "tail" of the distribution will be on the right. Also, the mean is to the right of the median, and the mode is to the left of the median.

The correct answer is: Most data values will fall to the left of the mean

Question 8

Not answered

Marked out of 1

A frequency distribution graph shows that the mean is to the left of the median, and the mode is to the right of the median. The distribution is most likely to be

Select one:

- Negatively skewed
- Abnormal distribution
- Symmetrical distribution
- Distribution cannot be determined
- Positively skewed

Your answer is incorrect.

In a negatively skewed distribution, most of the data values will fall to the right of the mean, while the tail will be on the left. Also, the mean is to the left of the median, and the mode is to the right of the median.

The correct answer is: Negatively skewed

Question 9

Not answered

Marked out of 1

Which of the following is true for a positively skewed distribution?

Select one:

- Mode < Median < Mean
- Mode = Median = Mean
- Mean < Median < Mode
- Median < Mode < Mean
- Mode cannot be calculated for skewed distributions

Your answer is incorrect.

Skew is the result of low-frequency categories with abnormally high or low values. By definition, positive skew is a result of the addition of abnormally high-value outliers - these outliers shift mean to a higher value. But as the mode is the most commonly occurring value, skew should not change the mode. Median changes more than the mode (but less than mean) because the middle value shifts somewhat when new categories are added at the extremes of the distribution (depends on the number of observations added).

The correct answer is: Mode < Median < Mean

Question **10**

Not answered

Marked out of 1

What type of distribution can have mean = 55; median = 52; mode = 41?

Select one:

- Positively skewed
- Normal
- Bimodal
- Negatively skewed
- Gaussian

Your answer is incorrect.

As mean>median>mode, this is positively skewed.

The correct answer is: Positively skewed

Question 11

Not answered

Marked out of 1

A study reported that patients with a substance use disorder had on average 5.2 emergency department visits per month (SD = 8.7), using a sample of 500 substance users from a hospital registry. Assuming a normal distribution for number of visits, If the same analysis is repeated using data from County substance users registry with a sample size of 60,000 which of the following will be seen

Select one:

- The standard deviation for the observed mean will be higher
- The observed mean will be closer to true population mean.
- The standard deviation for the observed mean will not change
- Due to an increased sample size, the average will become meaningless.
- The parameter distribution in the sample will become more skewed.

Your answer is incorrect.

A very large sample size will reduce the uncertainty in estimating the population mean (standard error of the mean). Thus, larger samples will more closely reflect the distribution of the underlying population. Provided that the measured parameter is normally distributed, the central limit theorem suggests that a reasonably large sample will have a distribution closer to this normal distribution. But if the distribution of a parameter in the population itself is skewed in some way, having a larger sample will not result in a normal distribution.

The correct answer is: The observed mean will be closer to true population mean.

Question **12**

Not answered

Marked out of 1

In a scale designed to measure the creativity of a population, it was found that the score obtained were normally distributed. If z is the difference between an individual's score and the mean expressed in units of standard deviation (standard score), then the number of people in the sample whose z will be between -1 and +1 will be

Select one:

- 68%
- 95%
- 100%
- 34%
- 2.3%

Your answer is incorrect.

In a standard normal distribution, 68% of the values fall in a range between -1 nads +1. This is same as ±1 standard deviations from the mean in a (nonstandard) Normal distribution.

The correct answer is: 68%

Question 13

Not answered

Marked out of 1

Which of the following is incorrect concerning skewed distributions?

Select one:

- In a negative skew, the mode will most likely be close to the shorter tail
- In a negative skew, the mean will most likely be close to the longer tail
- In a positive skew, the mode will most likely be close to the longer tail
- In a skewed distribution curve, one tail will be longer than the other
- In a positive skew, the mean will most likely be close to the longer tail

Your answer is incorrect.

In a positive skew, the shorter tail will be on the low-value side, (generally left) and the longer tail on the higher value side (positive, right) as there are some abnormally high observations. As mean shifts towards the right, the mode will be seen towards the shorter tail, i.e. the left side. In general, if the curve has one tail longer than the other (skewed), the mean is always toward the long tail, the mode is nearer the short tail, and the median is somewhere between the two.

The correct answer is: In a positive skew, the mode will most likely be close to the longer tail

Question **14**

Not answered

Marked out of 1

Among women aged 18 and above in a scholastic community, IQ is normally distributed with a mean of 100 and a standard deviation of 20. What percentage of women can be termed as geniuses with IQ more than 140?

Select one:

- 0.1
- 0.32
- 0.68
- 0.05
- 0.025

Your answer is incorrect.

The proportion that lie above two standard deviations is 2.5%.

The correct answer is: 0.025

Question **15**

Not answered

Marked out of 1

The number of times that a value occurs in a data set is called

Select one:

- Mode
- Median
- Range
- Frequency
- Mean

Your answer is incorrect.

The number of times a data occurs in a set is called the frequency. The mode is simply the most commonly occurring value in a frequency distribution. The highest peak of a frequency distribution curve represents the mode.

The correct answer is: Frequency

Question **16**

Not answered

Marked out of 1

A sample of the number of admissions to a psychiatric ward at a local hospital in South Wales during the full phases of the moon is given in the below image. How many observations have been made in total?

1	9
2	0 1 1 2 5 6 7 7 8
3	0 0 0 1 2 3 4 5 6 8
4	1 3

Select one:

- 22
- 9
- 10
- 4

- 19

Your answer is incorrect.

Simply count the number of observations in the leaf (not stem) of the plot. It is 22.

The correct answer is: 22

Question **17**

Not answered

Marked out of 1

A central point is chosen in a data set and deviation of individual values in either direction from this central value is calculated. It is found that the absolute value of deviations is similar on both sides. The central value is most likely to be

Select one:

- Standard deviation
- Variance
- Median
- Mode
- Mean

Your answer is incorrect.

The mean is simply the arithmetic average of all the items in a sample. To compute a sample mean, add up all the sample values and divide by the size of the sample.

The correct answer is: Mean

Question **18**

Not answered

Marked out of 1

A central point is chosen in a data set and number of individual values above or below this central value is calculated. It is found that the number of individual values greater than the central value is same as the number of values less than the central value. The central value is most likely to be

Select one:

- Mean
- Variance
- Mode
- Standard deviation
- Median

Your answer is incorrect.

The median is the middle value of an ordered frequency distribution so that roughly half of the data are smaller, and roughly half of the data are larger.

The correct answer is: Median

Question 19

Not answered

Marked out of 1

Which of the following is true regarding mode?

Select one:

- If two elements of a data set occur in the highest frequency then the average of the two is taken as the mode.
- If all the elements in a data set have the same frequency of occurrence then the data is said to be unimodal
- Mode will always be higher than the median irrespective of data distribution
- A frequency polygon of bimodal data shows two peaks
- All numerical data sets have at least one mode

Your answer is incorrect.

If all elements occur in same frequency the data will be multimodal; if there is one value that is outstanding as the most frequent, it defines an unimodal distribution.

The correct answer is: A frequency polygon of bimodal data shows two peaks

Question **20**

Not answered

Marked out of 1

Which of the following measures of central tendency is the most sensitive to change to any of the individual values in a data set?

Select one:

- Range
- Mean
- Interquartile range
- Median
- Mode

Your answer is incorrect.

The mean is very unstable. The mode is the most stable.

The correct answer is: Mean

Question 21

Not answered

Marked out of 1

If the same constant is added to all individual values in a data set, then the mean of the data is

Select one:

- Increased by the amount added
- Increased as a product of the constant value added to a dataset
- Reduced by the amount added
- Unchanged
- Reduced by twice the amount added

Your answer is incorrect.

The mean is increased by a positive value added to all observations.

The correct answer is: Increased by the amount added

Question 22

Not answered

Marked out of 1

While auditing CBT waiting list data, a trainee finds that five patients referred in last one week were given an appointment for assessment after 12, 13, 20, 15 and seven days. Which of the following would be affected the most if the trainee later finds a missed referral for which the first appointment was given 75 days later?

Select one:

- Frequency
- Mean
- Mode
- Median
- Interquartile range

Your answer is incorrect.

Adding an outlier to a set of observation disturbs the mean the most, followed by the median. The mode is the least disturbed measure of the three.

The correct answer is: Mean

Question 23

Not answered

Marked out of 1

While auditing CBT waiting list data, a trainee finds that five patients referred in last one week were given an appointment for assessment after 12, 13, 20, 15 and seven days. Which of the following is true regarding the central tendency of this dataset?

Select one:

- Dispersion cannot be determined as the number of observations is less than 10
- Median is the average of first and last observation in rank order
- The median, mode, and mean are equal.
- The median cannot be found for odd number of observations
- The data is multimodal

Your answer is incorrect.

A distribution may have any number of modes. Distributions with one mode are termed unimodal, those with two are termed bimodal. This data has no single value with higher frequency than others.

The correct answer is: The data is multimodal

Question 24

Not answered

Marked out of 1

To determine range of a dataset, we need to know

Select one:

- Mode and median
- Median and mean
- Largest and smallest values
- Largest value and standard deviation
- Smallest value and mode

Your answer is incorrect.

The spread or variability of the distribution is called the dispersion. A commonly used measure of dispersion is the range that is given by the difference between the highest and lowest values.

The correct answer is: Largest and smallest values

Question 25

Not answered

Marked out of 1

An approximate average of the squared deviations from the sample mean is called

Select one:

- Standard error
- 50th centile
- Standard deviation
- Variance
- Absolute deviation

Your answer is incorrect.

The variance is the measurement of the variation among all the subjects in an experiment. It is simply the square of the standard deviation. Like the standard deviation, the variance is a measure of the dispersion of the data set from the mean value. The variance is equal to the summation of the squared differences between each value and the mean, divided by the number of values in the dataset (usually n-1 which is the degree of freedom).

The correct answer is: Variance

Question 26

Not answered

Marked out of 1

It is felt that the average number of referrals for a regional mood disorder service is much more variable than the revenue generated from these referrals across a financial year. Which of the following statistical measures can be used to compare the variation of these two different variables?

Select one:

- Median values
- Standard deviation values
- Coefficient of variation
- Mean values
- Standard error values

Your answer is incorrect.

The coefficient of variation enables comparison of variations of two (or more) different variables. The coefficient of variation is truly meaningful only for values measured on a ratio scale.

The correct answer is: Coefficient of variation

Question 27

Not answered

Marked out of 1

Which of the following statements with respect to the standard deviation is false?

Select one:

- It has the same units as the original data
- If all observations have the same value then the sample standard deviation will be zero
- It includes all values in a dataset when calculated
- It cannot take a negative value
- It is not affected by outlier values

Your answer is incorrect.

The standard deviation is the most widely used measure of the spread of data about their mean. The larger the standard deviation, the more spread out the distribution of the data about the mean. The unit of SD is same as the unit of individual observations. If all of the data points are close to the mean, then the standard deviation will be small while if they are scattered or spread out over a large range of values, it will be large. Having high-value outliers will increase the standard deviation. It cannot take negative values as it is a square root of a fraction of summed squares. SD will be zero if all observations have the same value - almost never seen in clinical research.

The correct answer is: It is not affected by outlier values

Question **28**

Not answered

Marked out of 1

To study the influence of various diagnoses on the duration of hospital stay, an investigator calculates the ratio of standard deviation to the mean duration of stay for each diagnostic group observed during the period of study. This ratio is called

Select one:

- Standard error of the mean
- Hazard ratio
- Relative standard deviation
- Coefficient of variation
- Variance

Your answer is incorrect.

The coefficient of variation is defined as the sample standard deviation divided by the sample mean of the data set. This is generally expressed as a percentage. pb.rcpsych.org/cgi/reprint/14/1/1.pdf

The correct answer is: Coefficient of variation

Question 29

Not answered

Marked out of 1

Which of the following statements is correct with respect to a normal distribution?

Select one:

- Normal distribution is best suited for frequency counts
- No curve can be constructed
- Normal data cannot be summarised to mean values
- Central limit theorem is associated with sample size
- A smooth curve with biphasic peaks is formed

Your answer is incorrect.

When frequency measures of quantitative data are plotted, especially for continuous biological measures such as the number of people with various values of systolic BP, a smooth bell-shaped curve is often obtained. This is called normal or Gaussian distribution. It is often assumed that larger the sample, more normal the distribution of its means (central limit theorem). Central Limit Theorem states that for sample sizes sufficiently large the means will be normally distributed regardless of the shape of the original distribution. This is helpful because normally distributed data yields itself to more efficient ways of summarization.

The correct answer is: Central limit theorem is associated with sample size

Question 30

Not answered

Marked out of 1

With respect to mean, median and mode of a normally distributed data, which of the following assumption is correct?

Select one:

- Mean is greater than median
- Mean is smaller than both median and mode
- Mean lies between median and mode
- Only mean is calculated; median and mode cannot be ascertained.
- Mean is equal to median and mode

Your answer is incorrect.

Mean=median=mode for normal distribution.

The correct answer is: Mean is equal to median and mode

Question 31

Not answered

Marked out of 1

In a scale designed to measure the creativity of a population, it was found that the scores obtained were normally distributed. If mean score is 10 and standard deviation is 1, then the proportion of population who will score between 9 and 11 is

Select one:

- 0.34
- 0.68
- 0.95
- 1
- 0.023

Your answer is incorrect.

The SD = 1. The score 9 corresponds to Mean-1SD; 11 corresponds to mean+1SD. Hence, we can conclude that between -1SD and +1SD 68% of the population will lie if the data follows a normal distribution.

The correct answer is: 0.68

Question 32

Not answered

Marked out of 1

In an attempt to detect the stability of the diagnosis of binge eating disorder, the median lifetime duration of binge eating disorder was found to be ten years with an interquartile range of 3 to 22 years. Which of the following statement regarding the lifetime duration of binge eating disorder is correct?

Select one:

- The 50th percentile value is 3 years
- The 50th percentile value is 7 years
- In 25% of patients, the life time duration is less than 22 years
- The 75th percentile value is 22 years
- In 75% of patients , the life time duration varies between 3 and 22 years

Your answer is incorrect.

Here median = 50th percentile = 10. From interquartile range, we can detect 25th percentile is three years, and 75th percentile value is 22 years. Hence, 50% patients between 25th and 75th percentile will fall between 3 and 22 years.

The correct answer is: The 75th percentile value is 22 years

Question 33

Not answered

Marked out of 1

Which of the following is NOT a measure of dispersion?

Select one:

- Range
- Variance
- Mean
- Standard deviation
- Standard error

Your answer is incorrect.

The mean is a measure of central tendency.

The correct answer is: Mean

Question 34

Not answered

Marked out of 1

Which of the following is true about a negatively skewed distribution?

Select one:

- The mean is greater than the median
- Mean, mode and median are more or less equal.
- The tail is longer on left side
- The outliers are of extremely high value
- A smooth bell shaped symmetrical curve is seen

Your answer is incorrect.

In negative skew, lower value outliers lead to mean being smaller than the median and left tail being longer than right. Note that irrespective of the direction of a skew, the mode is always on the shorter side (outliers not as common as other values) and mean on the longer side.

The correct answer is: The tail is longer on left side

Question 35

Not answered

Marked out of 1

Standard deviation is more often used than variance while discussing the measure of dispersion of data. This is because

Select one:

- Variance can occur in negative values
- Standard deviation can have both positive and negative values
- Standard deviation has no units for expression
- Standard deviation is expressed in the same units as the primary data
- Variance is expressed in the same units as the primary data

Your answer is incorrect.

SD can only be positive as it is a square root of the sum of squares. The unit of expression of SD is same as the primary data. Variance is expressed in square units and not so meaningful.

The correct answer is: Standard deviation is expressed in the same units as the primary data

Question 36

Not answered

Marked out of 1

The number of days that a series of 5 patients had to wait before starting CBT at a centre is as follows: 12, 12, 14, 16, and 21. The median waiting time to get CBT is

Select one:

- 12 days
- 15 days
- 14 days
- 13 days
- 21 days

Your answer is incorrect.

Arrange this as 12,12,14,16,21. Median is central value = 14 days.

The correct answer is: 14 days

Question 37

Not answered

Marked out of 1

100 patients on clozapine had their blood pressure recorded soon after one dose of clozapine was given. The mean systolic BP was 150mmHg. If the standard deviation of this observation was 10, calculate the standard error of the mean blood pressure recorded in this sample.

Select one:

- 100
- 1.5
- 15
- 1.2
- 1

Your answer is incorrect.

Standard error is given by SEM = standard deviation/(sqr.root of sample size)

SEM = 10/SQRT100

SEM = 10/10 = 1

The correct answer is: 1

Question 38

Not answered

Marked out of 1

A lifestyle modification program was evaluated for its efficacy in reducing the excess weight gain seen in patients taking clozapine. Weight change (in kilograms) that occurred 6-months after the lifestyle modification program in 11 patients taking clozapine are shown in the table below in an order of magnitude. What is the mode?

Subject number	Weight change (before-after difference) in kilograms
1	4
2	5
3	5
4	5
5	6
6	7
7	8
8	9
9	9
10	10

11	17

Select one:

- 4
- 6
- 5
- 7
- 8

Your answer is incorrect.

A mode is the most commonly occurring observation. In this sequence, the value 5 Kg occurs three times, the highest frequency of observation.

The correct answer is: 5

Question 39

Not answered

Marked out of 1

A lifestyle modification program was evaluated for its efficacy in reducing the excess weight gain seen in patients taking clozapine. Weight change (in kilograms) that occurred 6-months after the lifestyle modification program in 11 patients taking clozapine are shown in the table below in an order of magnitude. What is the median change in weight?

Subject number	Weight change (before-after difference) in kilograms
1	4
2	5
3	5
4	5
5	6
6	7
7	8
8	9
9	9
10	10

11	17

Select one:

- 4
- 7
- 5
- 6
- 8

Your answer is incorrect.

The 50th percentile (median) is the value of the 6th person in this list of 11 subjects assembled by an order of magnitude (ascending order). The 6th person shows a change of 7 Kg. So 7 Kg is the median value.

The correct answer is: 7

Question 40

Not answered

Marked out of 1

A lifestyle modification program was evaluated for its efficacy in reducing the excess weight gain seen in patients taking clozapine. Weight change (in kilograms) that occurred 6-months after the lifestyle modification program in 11 patients taking clozapine are shown in the table below in an order of magnitude. What is the interquartile range of the weight difference?

Subject number	Weight change (before-after difference) in kilograms
1	4
2	5
3	5
4	5
5	6
6	7
7	8
8	9
9	9
10	10

11	17

Select one:

- 5
- 4
- 6
- 7
- 8

Your answer is incorrect.

We find the 25th percentile value as the value of 3rd person (5 Kg) and 75th percentile value as the value of the 9th person (9 Kg). The difference between these two values i.e. 9 - 5kg is 4Kg. So 4Kg is the interquartile range.

The correct answer is: 4

Question **41**

Not answered

Marked out of 1

Which of the following is true in negative skew?

Select one:

- mean is less than median.
- mean, mode median are the same
- mode is equal to mean
- mean is greater than median
- median is greater than mode

Your answer is incorrect.

The mean is larger than the median in positive skew

The correct answer is: mean is less than median.